



Appraisal of groundwater quality for suitability of drinking and irrigation purposes of pandameru river basin, anantapur district, AP, India

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Abstract

This study aimed to assess the quality of groundwater for drinking and irrigation in the Pandameru River basin, Andhra Pradesh, India. Groundwater samples were collected from 46 locations within the research region during the pre-monsoon season. These samples were analyzed for various parameters such as pH, EC, TDS, Ca^{2+} , Mg^{2+} , Na^+ , K^+ , HCO_3^- , Cl^- , SO_4^{2-} , and NO_3^- . The results were compared with WHO guidelines, indicating slightly alkaline groundwater. Analysis using the WQI and GIS-based interpolation provided an overall evaluation of water quality. WQI values range from 78.49 to 115.92 mg/L, indicating that 77% of the samples are safe for drinking, while 23% were unfit. Groundwater samples are found in the field of rock dominance, based on the Gibbs plot. The majority of the samples fell into the C3S1 category, which denotes water with high salinity and low sodium which is moderately suitable for irrigation purposes, according to the USSS diagram. Regarding cultivation, 78% were good, 20% doubtful, and 2% unsuitable based on the Wilcox classification. Except for a few locations, most samples were suitable for irrigation. However, some samples exceeded allowable contaminant levels due to human activity, requiring treatment before consumption.

Keywords Geographical information system (GIS) · Groundwater · Hydrogeochemistry · Irrigation water quality · Water quality index (WQI)

Introduction

Groundwater demand has recently increased significantly due to population development, urbanization, industrialization, and intensive farming practices in many regions of the world. Due to human actions, the quality of groundwater is changing, particularly for shallow aquifers (Adimalla et al. 2018; Sunitha and Reddy 2022). The deterioration of water

quality is extremely hazardous since human influence and environmental issues are increasing health concerns in many natural water bodies, such as rivers, lakes, and coastal areas. As a result, assessing groundwater resources is critical for ecological responsibility, control, and predicting prospective changes in climate. Man-made and geological processes are the two major elements that impact groundwater resources. Because of human activities, such as urbanization and the green revolution, the release of contaminated commercial and farming wastes into the environment upsets the biological equilibrium as technology advances (Subba Rao 2018). Numerous geogenic processes also affect groundwater resources. The quality varies within proximity in the research area because of highly localized factors including terrain and lithology (Singh et al. 2015; Rao and Latha 2019; Gaikwad et al. 2022). The following factors affect the quantity of the water, which includes weathering, subsoil flow, individual ion content and ion exchange, and environmental influences (Rajasekhar et al. 2019; Suvarna et al. 2022).

Groundwater quality in India is affected by urbanization, excessive extraction, fertilizer use, waste disposal, and

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geological factors. The water quality index (WQI) is a valuable tool for assessing water suitability, providing reliable information to the public (Ram et al. 2021). This approach monitors water quality, determining the suitability of drinking water in specific areas. Studies emphasize WQI's usefulness in evaluating overall water quality and assessing drinking water suitability (Kumar et al., 2022; Krupavathi et al. 2022). Researchers are developing water quality indices for groundwater investigations, selecting indices based on groundwater characteristics and intended outcomes. WQI presents groundwater information as a single value, indicating overall water quality through indicator numbers, expressing the impact of various water quality metrics (Etikala et al. 2022). Efforts are underway to compute WQI using physicochemical data from the Anantapur region, creating a crucial database for water capacity design and development.

A Geographic Information System (GIS) efficiently collects and connects vast amounts of geographically linked information, crucial for spatial analysis and processing. GIS proves valuable in managing local water supplies, addressing water resource challenges, analyzing groundwater resources, evaluating water quality, controlling flooding, and understanding ecosystems (Chandran et al. 2017; Chaurasia et al. 2018). Researchers have classified irrigation water quality using parameters like EC, SAR, % Na, RSC, and PI in various regions (Gowd 2005; Golla et al. 2021; Rishi et al. 2020; Patience et al. 2021; Golla and Badapalli 2021; Anusha et al. 2022). The study region, characterized by harsh terrain and insufficient surface water, relies heavily on groundwater for daily needs. While groundwater quality was better in the past (Aghazadeh et al. 2017; Etikala et al. 2019; Ahmed et al. 2021; Natesan et al. 2022; Golla et al. 2022), a current water quality assessment is essential.

There is a need to investigate groundwater quality for drinking and agricultural uses in the Pandameru River basin in south India. This research aimed to identify the physicochemical parameters of groundwater, evaluate its quality for drinking and agriculture using geospatial techniques, and consider potential sources of contamination. The study's results can inform science-based water health policies and provide valuable information for drinking and irrigation purposes in the Pandameru River basin.

The contribution of this study lies in its comprehensive assessment of groundwater quality, considering the specific context of the Pandameru River basin. By collecting groundwater samples from various locations within the basin during the appropriate pre-monsoon season, a thorough analysis of key parameters, such as pH, EC, TDS, Ca^{2+} , Mg^{2+} , Na^+ , K^+ , HCO_3^- , Cl^- , SO_4^{2-} , and NO_3^- , was conducted. This analysis allows for an accurate evaluation of the chemical composition of the groundwater and its potential impact on human health and agricultural activities. Moreover, this study goes beyond previous research by utilizing well-established water

quality guidelines, particularly those set by the World Health Organization (WHO), to assess the suitability of the groundwater for drinking purposes. The findings of this assessment contribute to a better understanding of the overall safety and quality of the groundwater in the Pandameru River basin, providing valuable information for policymakers, water resource managers, and local communities.

In addition, this study focuses on assessing groundwater suitability for irrigation purposes within the Pandameru River basin. Through the application of advanced techniques, such as the USSL diagram and the Wilcox classification, the research provides a deeper understanding of the water's salinity levels, sodium content, and potential implications for agricultural productivity. By identifying areas suitable for irrigation and areas requiring precautionary measures or alternative water sources, the study contributes to the promotion of sustainable agricultural practices in the region.

To summarize, this research article makes significant contributions through its comprehensive evaluation of groundwater quality for both drinking and irrigation purposes in the Pandameru River basin. By incorporating local data, adhering to international water quality guidelines, and utilizing advanced assessment techniques, the study provides valuable insights that can inform effective water resource management, facilitate informed decision-making, and guide the development of targeted interventions. Ultimately, the findings aim to ensure the availability of safe drinking water and promote sustainable agricultural practices in the region.

Organization of manuscript

The research article is organized into several key sections, including "Introduction" that outlines the significance and objectives of assessing groundwater quality in the Pandameru River basin. It incorporates a literature review to establish the research context. The "Methodology" section describes data collection and analysis techniques. The "Results and discussion" section presents the findings, followed by a discussion that interprets the results and relates them to existing knowledge. The "Conclusion" section summarizes the main findings and their implications, and the references list the cited sources.

Study area

The Pandameru River basin is a subbasin of the Pennar River. It covers an area of around 671 km². It lies between 14° 15' 00" to 14° 40' 00" N latitude and 77° 25' 00" to 77° 40' 00" E longitude in SOI toposheet No. 57 F/6, F/7, F/10, and F/11 at a scale of 1:50,000 in the southern portion of Anantapur district, Andhra Pradesh state, and a part of Tumkur district, Karnataka state (Fig. 1). Pandameru

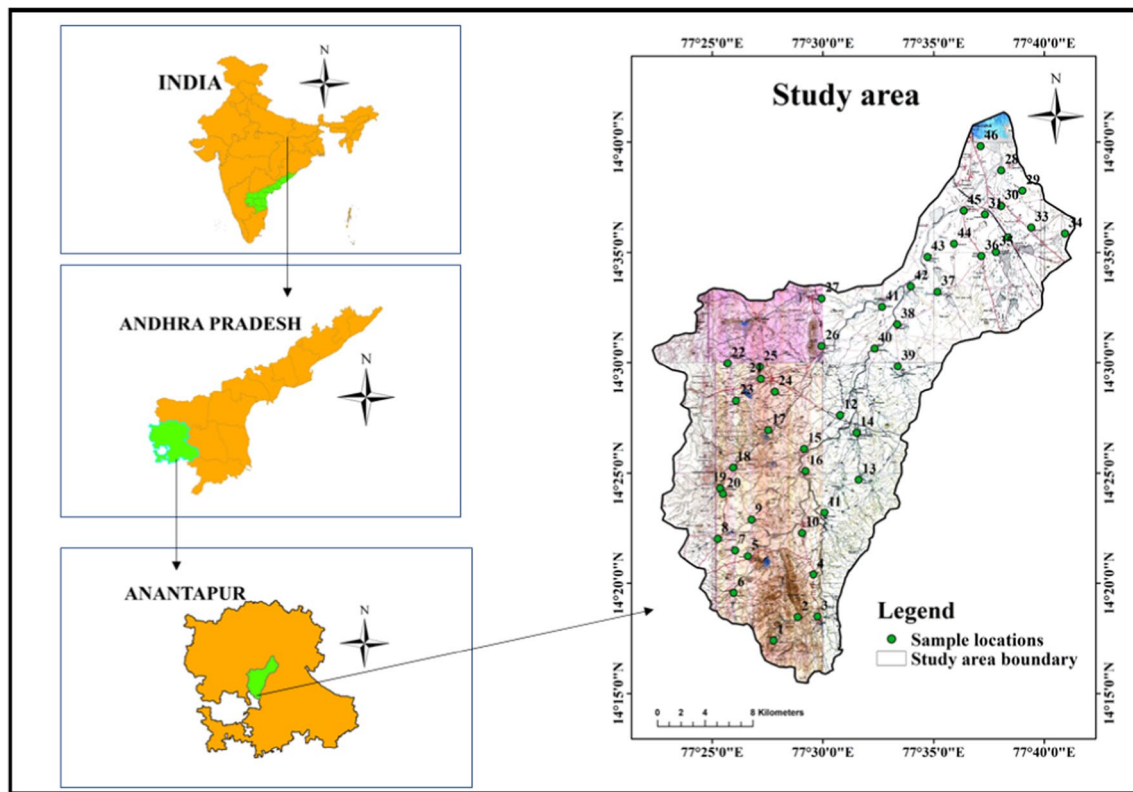


Fig. 1 Sample location map of the research region

River begins in Kangalakunta village in Ramagiri mandal and flows through Kanaganapalle, Raphadu, and Anantapur mandals before joining the Anantapur tank. The average yearly temperature ranges from 31 to 42 degrees Celsius. November, December, January, and February are the most pleasant months of the year. Temperatures can reach 42 °C throughout the summer. The temperature drops with the arrival of the southwest monsoon in early June. The temperature continues to fall gradually when the southwest monsoon departs in October (Badapalli et al., 2022). The mean annual precipitation in the research area is 594 mm. The lithology of the region includes rocks such as grey granite/pink granite, metabasalt, granite, granodiorite, hornblende biotite gneiss, biotite gneiss, and migmatite. Figure 2 depicts the lithological map of the research region.

Methodology

Sample collection

During the pre-monsoon season of March 2022, a total of 46 groundwater samples were carefully collected from various bore wells and hand pumps within the designated research region to conduct a comprehensive assessment of

groundwater quality. To ensure the accuracy of the results, 1-L capacity plastic bottles were utilized for sample collection. Prior to filling the bottles with groundwater samples, a rigorous cleansing process was carried out in the laboratory using diluted HNO₃ acid, followed by rinsing with distilled water. This meticulous procedure aimed to eliminate any potential contamination and maintain the integrity of the collected samples, ensuring reliable and accurate analysis.

Analysis

The American Public Health Association recommended analyzing groundwater samples for a variety of chemical characteristics (APHA 1995). Using a portable conductivity meter, pH and EC were measured in the field. EC was multiplied by 0.64 to calculate the total dissolved solids (TDS) (Hem 1985). The volumetric approach was used to examine TH and TA as CaCO₃, Ca²⁺, CO₃²⁻, HCO₃⁻, and Cl. Mg²⁺ was estimated using the contents of TH and Ca²⁺. Sulfate and nitrate concentrations were measured for this analytical investigation using a PerkinElmer Lambda 25 UV–Visible Spectrophotometer. In this investigation, sodium and potassium concentrations in groundwater samples were determined using the ESICO Digital Clinical Flame Photometer. By associating the values of several water excellence

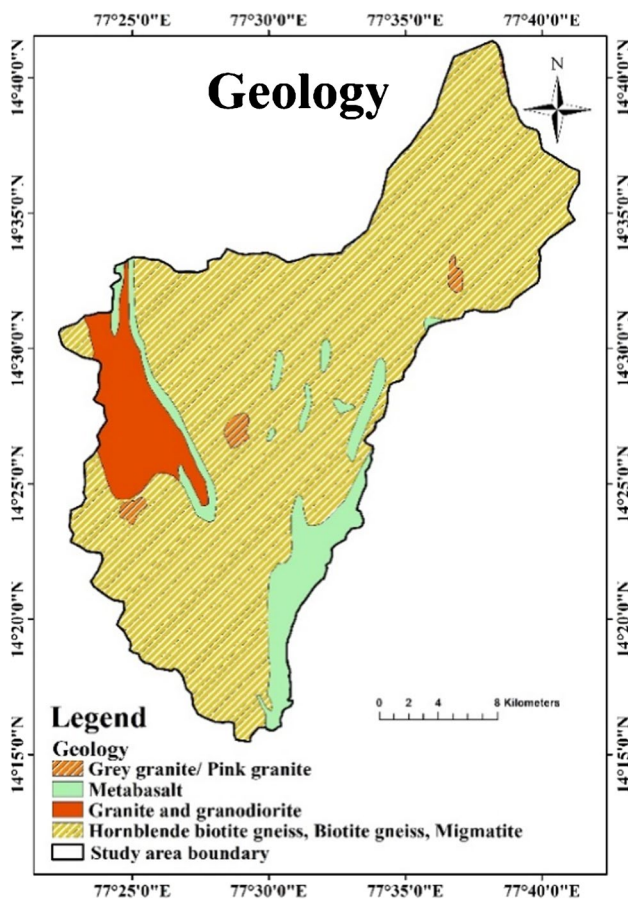


Fig. 2 Lithological map of the research region

parameters to the World Health Organization 2012 drinking water standard values, the acceptability of the groundwater for consumption, business, residential, and farming applications was assessed.

GIS analysis

The Survey of India topographic sheets 56F/6, 57F/7, 57F/10, and 57F/11 were used to create the basic map for the research area using the ArcGIS 10.4 program (Fig. 1). Each sampling site was located using German GPS, and the coordinates were then entered into a GIS platform for further research. The inverse distance weighted interpolation (IDW) method was utilized to accomplish this (Nath et al., 2021; Kumar et al., 2019). IDW is a technique for interpolating values between observations that may be calculated spatially. Each value represents the weighted average of the neighboring locations. Weights are calculated using the inverse distance method from the location observation to the estimation point. The spatial analyst software extension is used to interpolate the IDW technique, which is a valuable tool for creating spatial variation maps of the subsurface

physicochemical properties of these locations (Subba Rao 2018). The IDW technique has been widely used around the world for a variety of reasons, resulting in spatial distribution maps that distinguish concentration zones significantly.

Anion-cation balance

Because the water in a natural environment is electrically neutral, the sum of all anions' concentrations should equal the sum of all cations' concentrations, with values reported in meq/L. This analysis is based on the percent difference (percent diff) value calculated in the equation below. All of the groundwater samples are under the approved $\pm 5\%$ level.

$$\text{Charge - Balance Error (CBE)} = \frac{\sum \text{Cations} - \sum \text{anions}}{\sum \text{Cations} + \sum \text{anions}} \times 100$$

Water quality index

The significance of various parameters in WQI formulation depends on the intended use of water. It reduces a large quantity of data about water superiority to a single numerical value. Water quality parameters are studied in terms of acceptability for human consumption. The standards utilized in this study are those recommended by World Health Organization (2012) for the various drinking water parameters. As mentioned before, the WQI was calculated using eleven different water quality parameters. As mentioned before, eleven water quality parameters were considered for calculating the WQI. The water quality index (WQI) uses the World Health Organization (WHO) drinking water quality standards. As shown in the following steps, WQI was calculated using a weighted arithmetic index method (Brown et al., 1970).

Calculation of water quality (Q_n)

$$Q_n = 100 \left[(V_n - V_i) / (S_n - V_i) \right]$$

where

Q_n = n th parameter of water quality rating.

V_n = the value of the n th parameter observed,

S_n = standard acceptable value of the n th parameter,

V_i = ideal value for the n th parameter.

All ideal values (V_i) are taken as zero for drinking water except pH 7.0.

Unit weight calculation (W_n)

The unit weight of the corresponding parameter was calculated by an inversely proportional value to the recommended standard value S_n .

$$W_n = K/S_n$$

where

W_n = n th parameter of unit weight,

S_n = n th parameter standard value,

K = constant proportionality.

The proportionality constant (K) has been calculated here using the equation:

$$K = 1 / \sum (1/S_n)$$

The total water quality index (WQI) was then calculated linearly by adding the quality assessment to the unit weight.

$$WQI = \sum W_n Q_n / \sum W_n$$

where

Q_n = quality rating.

W_n = relative weight.

Results and discussion

Water appropriateness for several drives was decided by its quality, which was based on certain standards. Water for drinking must be soft, low in dissolved solids, and devoid of dangerous pollutants (Fuchu et al. 2021). The expressive data of the studied parameters are supplied in Table 1. The results are associated with the maximum allowable limits set by the World Health Organization (2012).

The pH of the groundwater in the research area ranged from 7.4 to 8.3, indicating a moderately alkaline nature (Table 1 and 2). A pH range of 6.5–8.5 is appropriate for drinking and agricultural purposes (ISI 2012). All of the samples are within the allowed limits. The EC values of groundwater samples ranged from 870 to 3580 s/cm at 25 °C (Table 1 and 2). The concentration of TDS in groundwater varies between 5 and 1790 mg/L. TDS concentration is

largely determined by the ions present in water. Freeze and Cherry (1979) classified water into four types based on TDS content. According to this categorization, 36 groundwater samples are classified as fresh, while the remaining 10 are classified as brackish (Table 7). The concentration of TH in groundwater ranges between 286 and 1043 mg/L (Table 1 and 2). The TH (Sawyer and McCarty 1967) category is shown in Table 7, and the bulk of the groundwater samples fall into the extremely hard group, making them unfit for consumption.

Cation anion chemistry

The principal cations are listed below in order of reducing concentration given in mg/L: $\text{Ca}^{2+} > \text{Na}^+ > \text{Mg}^{2+} > \text{K}^+$. According to the analytical results, the concentration of Ca^{2+} in the research area varies from 59 to 258 mg/L, with a mean of 124.4 mg/L (Table 1 and 2). Ca^{2+} has a maximum permissible value of 200 mg/L (World Health Organization 2012). The higher calcium concentration in Nimalivaram could be related to the dissolution of CaCO_3 and $\text{Ca Mg}(\text{CO}_3)_2$ precipitates during recharging. The analytical results revealed that the Mg^{2+} content varied from 9 to 167 mg/L, with a mean of 52.7 mg/L (Table 1 and 2). Mg^{2+} has a maximum permitted limit of 150 mg/L. Na^+ levels in the research area vary from 76 to 186 mg/L, with a mean of 112.2 mg/L. The higher sodium concentration in Nasanakota village could be related to the silicate weathering process and agricultural sources. The potassium (K^+) concentrations in groundwater samples vary from 1 to 27 mg/L, with an average of 7 mg/L (Table 1 and 2). The bulk of the samples fall under the WHO-recommended level of 12 mg/L. Four percent of all groundwater samples are found to be higher above the permitted level of 12 mg/L potassium. K^+ levels in groundwater may have risen as a result of the overuse of potassium in agriculture.

HCO_3^- levels in the research area vary from 349 to 748 mg/L with an average value of 517.5 mg/L (Table 1 and 2). Higher concentrations of HCO_3^- may be found in Narasampalli village due to the impact of CO_2 on the fundamental components of soil and granitic rock. SO_4^{2-} levels in the research region vary from 37 to 411 mg/L, with a mean of 111 mg/L (Table 1 and 2). Cl^- is primarily derived from a variety of sources, including household trash, manufacturing wastes, urban effluents, and rock weathering, and is used as a pollution indicator (Chaudhary and Satheeshkumar 2018). Cl^- levels in the research region varied from 89 to 284 mg/L, with a mean of 181 mg/L; 4% of samples exceeded permissible limits (World Health Organization 2012). One of the most significant nutrients in an environment is nitrate. Nitrate levels are generally greater in water bodies that have been contaminated by organic materials (Naaz 2015; Reddy et al. 2022).

Table 1 Analytical data of physicochemical parameters of the Pandameru River basin

Sample no	Sample locations	pH	EC ($\mu\text{S}/\text{cm}$)	TDS (mg/L)	Ca^{2+} (mg/L)	Mg^{2+} (mg/L)	Na^+ (mg/L)	K^+ (mg/L)	HCO_3^- (mg/L)	Cl^- (mg/L)	SO_4^{2-} (mg/L)	NO_3^- (mg/L)	TH (mg/L)
1	Kangalakunta	7.9	1580	790	117	9	126	7	488	114	98	10	329
2	Konchikunta	7.9	1620	810	216	12	121	10	610	130	112	0	589
3	Ramagiri	7.8	970	485	174	88	112	5	564	106	411	9	796
4	R. Kothapalli	8	1850	925	136	24	144	6	427	89	271	0	438
5	Nasanakota	7.7	1850	925	95	34	186	3	541	124	93	0	377
6	Muthyalanpalli	7.9	1470	735	127	28	138	3	428	175	129	0	432
7	Venkatapuram	7.6	1680	840	148	21	98	14	449	124	157	0	456
8	Kotta Gyadhikunta	8	1940	970	145	66	89	6	540	213	87	0	633
9	Madapuram	8.2	2360	1180	129	30	104	8	427	238	59	0	445
10	Muthavakuntla	8	1470	735	113	44	106	10	549	148	116	0	463
11	Thallimadugula	8	1910	955	80	42	112	1	488	142	49	0	372
12	Balepalyam	8	2350	1175	185	22	92	11	549	213	37	22	553
13	Konapuram	7.9	1680	840	131	99	108	14	549	266	67	5	733
14	Kanaganapalle	7.9	2240	1120	166	97	102	7	671	190	107	34	813
15	Ellakuntla	7.7	920	460	151	112	87	27	660	240	79	0	837
16	Nemalivaram	7.9	870	435	258	59	104	7	710	228	89	35	887
17	Bhadhalapuram	7.8	1580	790	124	91	82	4	610	177	102	0	683
18	Narasampalli	7.8	2310	1155	240	108	94	6	748	243	164	22	1043
19	Somavarand-lapalli	7.8	1880	940	193	88	132	4	732	213	128	0	843
20	Gudivandlapalli	8	2450	1225	184	78	129	5	650	218	127	0	780
21	Thagarakunta	8	1470	735	96	46	142	4	419	239	79	0	429
22	Guntapalli	8	1980	990	69	41	137	2	428	195	55	0	341
23	Patapalem	7.6	1550	775	74	43	138	2	428	213	48	0	361
24	Pullaikunta	7.8	1670	835	117	74	97	4	488	180	80	0	596
25	Boginayanipalli	8.2	910	455	120	60	87	3	488	142	136	0	546
26	Palacherla	8.3	2360	1180	175	53	104	6	549	284	90	0	655
27	Thopudhurthi	7.7	1770	885	185	101	94	4	640	228	134	10	877
28	Upparapalle	7.6	1690	845	116	31	116	10	448	177	126	0	417
29	Chinnakuntapalli	7.4	890	445	112	20	122	6	419	182	96	0	362
30	Akutotapalli	7.9	1880	940	84	39	128	3	469	175	77	9	370
31	Gangalakunta	7.5	1360	680	108	66	118	10	489	155	250	20	541
32	Kandukururu	7.4	1080	540	115	69	84	8	549	186	68	0	570
33	Itukalapalle	7.8	1680	840	89	59	98	12	529	157	91	5	464

Table 1 (continued)

Sample no	Sample locations	pH	EC (µS/cm)	TDS (mg/L)	Ca ²⁺ (mg/L)	Mg ²⁺ (mg/L)	Na ⁺ (mg/L)	K ⁺ (mg/L)	HCO ₃ ⁻ (mg/L)	Cl ⁻ (mg/L)	SO ₄ ²⁻ (mg/L)	NO ₃ ⁻ (mg/L)	TH (mg/L)
34	Krishnamareddipalle	7.8	1780	890	84	37	116	10	489	142	78	8	362
35	zangalapalle	7.8	2450	1225	109	24	111	4	349	142	151	0	371
36	Ramanepalli	8	3580	1790	96	30	112	10	478	146	73	10	363
37	Gondireddypalle	8	2550	1275	65	30	136	7	428	148	87	0	286
38	Erragunta	8.1	1980	990	67	67	91	7	449	122	192	10	442
39	Bandameedapalle	7.8	1710	855	101	101	108	8	732	160	105	0	667
40	Varimadugu	7.5	1640	820	126	53	125	5	488	213	79	0	532
41	Gandlaparthi	8.1	2150	1075	76	56	107	3	427	195	97	0	420
42	Pullalarevu	8.2	1660	830	126	31	103	10	448	203	87	0	442
43	Bommaparthi	8.1	1800	900	59	167	76	5	549	249	106	0	832
44	Lingampalli	7.9	1360	680	88	40	108	9	428	193	72	8	384
45	Rapthadu	7.6	980	490	83	32	112	9	379	172	88	7	339
46	Anantapur tank	7.8	1890	945	69	39	117	5	429	145	78	10	332

Ca²⁺, Mg²⁺, Na⁺, K⁺, HCO₃⁻, SO₄²⁻, Cl⁻, TDS, and TH are expressed in mg/L, EC is in µS/cm, pH units

Thirty-five milligrams per liter of nitrate was the highest amount observed in the study region. According to the World Health Organization 2012 guidelines, all water samples had low nitrate concentrations and were below the allowed values (45 mg/L).

Water quality index (WQI)

The water quality index of relative weights for each parameter is given in Table 3, and WQI at individual sampling locations is shown in Table 5. The computed WQI values in this study vary from 78.49 to 115.92. As a result, this can be divided into five categories: Excellent, Class I (50); Good, Class II (51–100); Poor, Class III (101–200); Very poor, Class IV (201–300); Unsuitable for drinking, Class V (> 300) (Table 4). According to the water quality index classification, 77% of the groundwater samples fall into Class II: good, while the remaining 23% fall into Class III: bad water, as indicated in Table 4. The high value of WQI may be related to greater amounts of TDS, EC, and TH. These findings suggest that the groundwater is just slightly suitable for consumption (Table 5). Figure 3 shows the spatial distribution map of WQI.

Irrigation purpose

The appropriateness of groundwater for irrigation is determined by the impact of inorganic elements on the aquatic on both soil and vegetation (Wilcox 1955; Richards 1954). Drainage is a key component that links agricultural development to water excellence. Even if the soil has a lot of saline water, crops can grow on it if it is effectively drained. Conversely, yields may not be adequate in a region with poor drainage and high-quality water (Todd and Mays 2004). Some essential hydrochemical measures such as EC, %Na, SAR, RSC, PI, KI, and MR play a critical role in assessing irrigation water quality. EC and SAR are well-known as good indicators of salinity and sodium toxicity to crops, respectively (Table 6). When SAR is displayed versus EC (Fig. 4), it is clear that 83% of the research region's groundwater falls into the C3S1 category, which denotes water with high salinity and low brackish content, making it ideal for agriculture in practically all soil types (USSL 1954; Table 7). However, 17% of groundwater samples are classified as C4S1, suggesting a type with very high salt concentration and little sodium that is unsuitable for agriculture. Table 6 displays the results. The sodium adsorption ratio (SAR) is also frequently used to determine groundwater suitability for irrigation (Karanth 1989), and it is computed as follows:

$$SAR = \frac{Na^+}{\sqrt{\frac{(Ca^{2+} + Mg^{2+})}{2}}}$$

Table 2 Statistical analysis and comparison of groundwater quality parameters with WHO 2012 criteria

Parameter	Minimum	Maximum	Average	WHO (2012)		BIS (2012)	
				Desirable limit	Permissible limit	Desirable limit	Permissible limit
pH	7.4	8.3	7.9	6.5–8.5	9.2	6.5	8.5
EC	870	3580	1756.5	500	1500	–	–
TDS	435	1790	878.3	500	1500	500	2000
Ca ²⁺	59	258	124.4	75	200	–	–
Mg ²⁺	9	167	55.7	50	150	30	100
Na ⁺	76	186	112.2	50	200	200	400
K ⁺	1	27	7	–	12	–	–
HCO ₃ [–]	349	748	517.5	–	–	–	–
Cl [–]	89	284	181.2	250	600	250	1000
SO ₄ ^{2–}	37	411	111	250	400	200	400
NO ₃ [–]	0	35	5.1	–	45	45	No relaxation
TH	285.5	1042.8	539.2	100	500	200	600

Table 3 Relative weight of each parameter

Chemical parameters	S _n (World Health Organization 2012)	1 / S _n	Σ1 / S _n	K	W _n = K / S _n	Ideal value (V _i)	Observed value (V _n)	V _n / S _n	V _n / S _n × 100 = W _n	W _n Q _n	WQI
pH	8.5	0.118	0.194	5.1493	0.605795	7	7.9	0.93	92.9	56.3	81.34
EC	500	0.002	0.194	5.1493	0.010299	0	1580	3.16	316	3.25	
TDS	500	0.002	0.194	5.1493	0.010299	0	790	1.58	158	1.63	
Ca ²⁺	75	0.013	0.194	5.1493	0.068657	0	117	1.56	156	10.71	
Mg ²⁺	50	0.02	0.194	5.1493	0.102985	0	9	0.18	18	1.85	
Na ⁺	200	0.005	0.194	5.1493	0.025746	0	126	0.63	63	1.62	
HCO ₃ [–]	500	0.002	0.194	5.1493	0.010299	0	488	0.98	97.6	1.01	
Cl [–]	250	0.004	0.194	5.1493	0.020597	0	114	0.46	45.6	0.94	
SO ₄ ^{2–}	250	0.004	0.194	5.1493	0.020597	0	98	0.39	39.2	0.81	
NO ₃ [–]	45	0.022	0.194	5.1493	0.114428	0	10	0.22	22.2	2.54	
TH	500	0.002	0.194	5.1493	0.010299	0	329	0.66	65.8	0.68	
		0.194								81.34	

SAR values were categorized into four categories by Richards (1954) based on the levels of Na⁺, Ca²⁺, and Mg²⁺ in meq/L (Table 6). The SAR values in the current research vary from 1.14 to 3.50, showing that the water in the study region has excellent irrigation quality (Table 6).

Table 4 Classification of water quality types based on WQI

WQI range	Class	Water quality status	% of water samples
< 50	I	Excellent water	–
51–100	II	Good water	77%
101–200	III	Poor water	23%
201–300	IV	Very poor water	–
> 300	V	Unsuitable water	–

Percentage sodium ((%Na)

Sodium content is another important component to consider when categorizing irrigation water, even though sodium interacts with loam to impact soil porousness and consistency. Additionally, sodium content is frequently represented as a percentage of sodium (%Na), which is derived following Wilcox's (1955) formula:

$$\%Na = \frac{Na^+ + K^+}{(Ca^{2+} + Mg^{2+} + Na^+ + K^+)} \times 100$$

The % Na in the groundwater in the studied region ranges from 16.90 to 51.63%. About 68% of collected groundwater samples are within permissible irrigation limits, 7% are in the good category for irrigation, and the remaining 25% are in the undecided class for irrigation (Table 7), because

Table 5 WQI at individual sampling locations

Sample no	Sample locations	Latitude	Longitude	WQI	Water quality status
1	Kangalakunta	N 14° 17' 24.63"	E 77° 27' 45.42"	81.34	Good water
2	Konchikunta	N 14° 18' 27.94"	E 77° 28' 51.38"	89.57	Good water
3	Ramagiri	N 14° 18' 29.47"	E 77° 29' 45.79"	103	Poor water
4	R. Kothapalli	N 14° 20' 23.37"	E 77° 29' 34.39"	86.73	Good water
5	Nasanakota	N 14° 21' 13.96"	E 77° 26' 36.53"	81.98	Good water
6	Muthyalanpalli	N 14° 19' 34.09"	E 77° 25' 55.19"	84.29	Good water
7	Venkatapuram	N 14° 21' 29.75"	E 77° 26' 01.95"	82.67	Good water
8	Kotta Gyadhikunta	N 14° 22' 00.84"	E 77° 25' 15.14"	95.91	Good water
9	Madapuram	N 14° 22' 53.16"	E 77° 26' 46.23"	89.39	Good water
10	Muthavakuntla	N 14° 22' 16.94"	E 77° 29' 03.03"	86.64	Good water
11	Thallimadugula	N 14° 23' 12.42"	E 77° 30' 04.9"	83.68	Good water
12	Balepalyam	N14° 24' 22.90"	E 77° 30' 47.16"	96.85	Good water
13	Konapuram	N 14° 31' 37.14"	E 77° 31' 37.14"	101.56	Poor water
14	Kanaganapalle	N 14° 26' 49.35"	E 77° 29' 31.10"	113.74	Poor water
15	Ellakuntla	N 14° 26' 06.08"	E 77° 29' 08.62"	101.09	Poor water
16	Nemalivaram	N 14° 25' 04.89"	E 77° 29' 12.50"	107.99	Poor water
17	Bhadhalapuram	N 14° 26' 57.06"	E 77° 27' 31.85"	96.58	Good water
18	Narasampalli	N14° 25' 15.04"	E 77° 25' 56.46"	115.92	Poor water
19	Somavarandlapalli	N 14° 24' 18.85"	E 77° 25' 21.59"	105.46	Poor water
20	Gudivandlapalli	N 14° 24' 03.56"	E 77° 25' 29.55"	104.53	Poor water
21	Thagarakunta	N 14° 29' 15.54"	E 77° 27' 11.20"	86.01	Good water
22	Guntapalli	N14° 29' 57.89"	E 77° 27' 42.12"	83.3	Good water
23	Patapalem	N14° 28' 16.49"	E 77° 26' 04.56"	80.14	Good water
24	Pullaikunta Tanda	N14° 28' 40.60"	E 77° 27' 49.57"	92.16	Good water
25	Boginayanipalli	N14° 29' 48.72"	E 77° 29' 08.63"	90.14	Good water
26	Palacherla	N14° 30' 44.92"	E 77° 29' 56.61"	100.28	Poor water
27	Thopudhurthi	N14° 32' 54.46"	E 77° 29' 27.62"	107.55	Poor water
28	Upparapalle	N14° 38' 43.78"	E 77° 38' 03.14"	81.95	Good water
29	Chinnakuntapalli	N14° 37' 47.69"	E 77° 39' 01.01"	75.25	Good water
30	Akutotapalli	N14° 37' 06.37"	E 77° 38' 03.31"	85.19	Good water
31	Gangalakunta	N14° 36' 43.48"	E 77° 37' 19.77"	93.04	Good water
32	Kandukuru	N14° 35' 40.18"	E 77° 38' 22.31"	86.29	Good water
33	Itukalapalle	N14° 36' 07.69"	E 77° 39' 24.49"	87.57	Good water
34	Krishnamareddipalle	N14° 35' 51.25"	E 77° 40' 56.03"	83.17	Good water
35	zangalapalle	N14° 35' 01.20"	E 77° 37' 48.78"	83.41	Good water
36	Ramanepalli	N14° 34' 50.27"	E 77° 37' 09.76"	90.33	Good water
37	Gondireddypalle	N14° 33' 12.47"	E 77° 35' 10.46"	80.05	Good water
38	Erragunta	N14° 31' 43.82"	E 77° 33' 21.43"	91.92	Good water
39	Bandameedapalle	N14° 29' 49.86"	E 77° 33' 22.67"	97.38	Good water
40	Varimadugu	N14° 30' 38.27"	E 77° 32' 19.90"	87.09	Good water
41	Gandlaparthi	N14° 32' 31.74"	E 77° 32' 40.13"	88.26	Good water
42	Pullalarevu	N14° 33' 27.68"	E 77° 33' 57.76"	86.82	Good water
43	Bommaparthi	N14° 34' 47.96"	E 77° 34' 43.55"	109.46	Poor water
44	Lingampalli	N14° 35' 23.44"	E 77° 35' 55.42"	83.82	Good water
45	Rapthadu	N14° 36' 53.83"	E 77° 36' 21.93"	78.49	Good Water
46	Anantapur tank	N14° 41' 46.20"	E 77° 36' 40.11"	83.23	Good Water

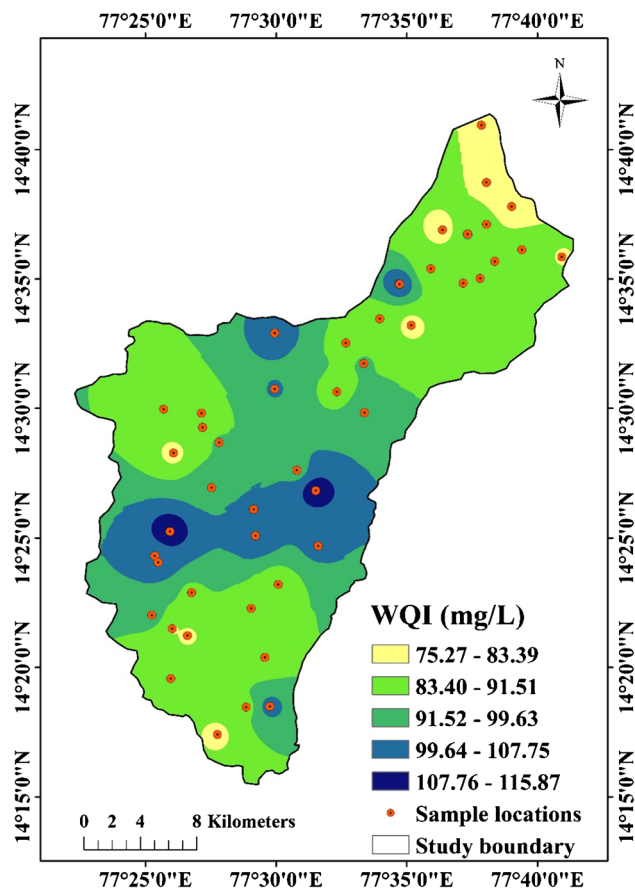


Fig. 3 Spatial distribution map of WQI

they reason deflocculation and damage of soil farmland and porousness (Karanth 1989).

Wilcox diagram

According to the Wilcox diagram, 78% of groundwater samples are good to permitted, 20% are doubtful to unsuitable, and only 2% are inappropriate for irrigation (Fig. 5).

Residual sodium carbonate (RSC)

RSC in the water shows that the HCO_3^- value is quite high in comparison to other negative ions (Table 6). As a result, carbonate ions ($\text{HCO}_3^- + \text{CO}_3^{2-}$) impact water quality by precipitating alkaline earth ($\text{Ca}^{2+} + \text{Mg}^{2+}$) in the soil, where the water is additionally intense (Tiwari et al. 2017). As a result, estimating RSC is critical for irrigation suitability and can be done using the following equation (Eaton 1950):

$$\text{RSC} = (\text{CO}_3^{2-} + \text{HCO}_3^-) - \text{Ca}^{2+} + \text{Mg}^{2+}$$

This displays each ionic level in meq/L. The observed RSC levels in groundwater were 8.66 to 1.38 meq/L. As indicated in Table 7, almost 93% of groundwater samples that were taken are within permissible ranges, whereas only 7% of water samples are possibly used for farming.

Permeability index (PI)

The permeability index is another significant factor that should be taken when judging if water is suitable for farming. The continuity formula (Doneen 1964) describes how sodium, calcium, magnesium, and bicarbonate concentrations affect infiltration capacity:

$$\text{PI} = \frac{\text{Na}^+ + \sqrt{\text{HCO}_3^-}}{(\text{Ca}^{2+} + \text{Mg}^{2+} + \text{Na}^+)} \times 100$$

The computed investigative data show that all of the groundwater samples were categorized as Class II, indicating that the groundwater is appropriate for farming (Table 7).

Chloralkaline index (CAI)

Schoeller (1977) discovered that groundwater's chemical composition changes over time and that its movement can be described by the chloralkaline indices CAI I and CAI II. Na^+ , K^+ , and Ca^{2+} from the water and Mg^{2+} and Ca^{2+} from the rocks directly exchanged ions because of the presence of a positive chloralkaline guide. Mg^{2+} and Ca^{2+} from water and Na^+ and K^+ from rocks exchange ions when CAI is negative. CAI I's resulting value was between -1.56 and 0.51 , whereas CAI II's was between -0.31 and 0.32 (Table 6). These figures showed that 52% of the analyzed region fell into the positive group, whereas 48% fell into the negative category (Table 7). The data clearly demonstrated to us that exchangeable cation can also be utilized to determine the chemical makeup of substances.

$$\text{CAI I} = \text{Cl}^- - \text{Na}^+ + \text{K}^+ / \text{Cl}^-$$

$$\text{CAI II} = \text{Cl}^- - (\text{Na}^+ + \text{K}^+) / \text{SO}_4^{2-} + \text{HCO}_3^- + \text{CO}_3^{2-} + \text{NO}_3^-$$

Kelley's ratio (KR)

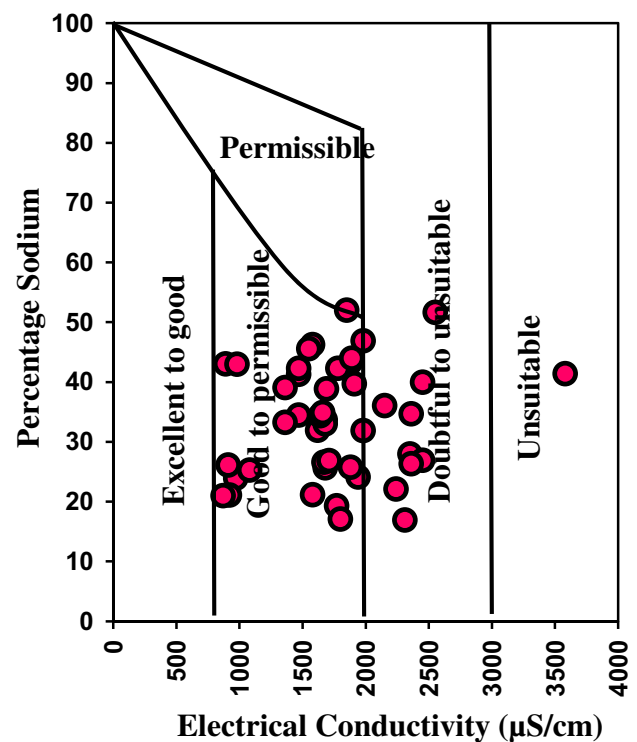
KR, which seems to be a measure of Na^+ to ca and mg, was used to categorize the irrigated agriculture aquatic environment (Kelley 1940). To calculate Kelley's ratio, use the following formula:

Table 6 Summary of hydrogeophysical parameters in groundwater

Sample no	SAR (meq/L)	RSC (meq/L)	PI (meq/L)	KR (meq/L)	MAAR (meq/L)	Indices of base exchange	Gibbs ratio I (meq/L)	Gibbs ratio II (meq/L)	%Na (meq/L)
						CAI-I (meq/L)	CAI-II (meq/L)		
1	3.03	1.38	68.84	0.83	11.25	-0.76	-0.24	0.29	0.48
2	2.17	-1.82	49.41	0.45	8.38	-0.51	-0.15	0.27	0.33
3	1.73	-6.72	38.01	0.31	45.45	-0.67	-0.11	0.25	0.36
4	3	-1.8	59.27	0.72	22.52	-1.56	-0.31	0.27	0.48
5	4.17	1.28	70.78	1.07	37.09	-1.34	-0.43	0.28	0.49
6	2.89	-1.66	59.03	0.69	26.64	-0.23	-0.12	0.41	0.49
7	2	-1.79	52.14	0.47	18.95	-0.32	-0.11	0.32	0.37
8	1.54	-3.86	41.38	0.31	42.85	0.33	0.19	0.41	0.35
9	2.14	-1.94	53.35	0.51	27.7	0.3	0.24	0.49	0.41
10	2.14	-0.31	54.82	0.5	39.08	-0.17	-0.06	0.32	0.45
11	2.52	0.51	62.45	0.65	46.38	-0.22	-0.1	0.33	0.55
12	1.7	-2.09	46.48	0.36	16.38	0.29	0.18	0.4	0.3
13	1.73	-5.73	39.68	0.32	55.45	0.33	0.24	0.46	0.42
14	1.56	-5.32	37.44	0.27	49.05	0.14	0.06	0.33	0.35
15	1.31	-5.98	34.42	0.23	54.99	0.34	0.19	0.39	0.33
16	1.52	-6.15	35.62	0.26	27.36	0.27	0.13	0.36	0.26
17	1.37	-3.72	38.97	0.26	54.73	0.26	0.11	0.33	0.37
18	1.27	-8.66	30.39	0.2	42.57	0.38	0.17	0.36	0.25
19	1.98	-4.93	40.66	0.34	42.89	0.03	0.01	0.33	0.37
20	2.01	-5	41.79	0.36	41.12	0.07	0.03	0.37	0.38
21	2.98	-1.74	59.58	0.72	44.11	0.07	0.05	0.5	0.56
22	3.22	0.16	67.33	0.87	49.47	-0.09	-0.06	0.44	0.63
23	3.16	-0.25	65.33	0.83	48.91	-0.01	-0.01	0.46	0.62
24	1.73	-3.97	43.61	0.35	51.03	0.15	0.08	0.39	0.42
25	1.62	-2.97	44.91	0.35	45.16	0.04	0.01	0.33	0.39
26	1.77	-4.14	42.66	0.35	33.28	0.42	0.31	0.47	0.34
27	1.38	-7.1	33.85	0.23	47.35	0.35	0.17	0.38	0.31
28	2.47	-1.03	57.88	0.61	30.57	-0.06	-0.03	0.41	0.47
29	2.79	-0.4	63.13	0.73	22.73	-0.06	-0.04	0.43	0.49
30	2.9	0.25	64.23	0.75	43.34	-0.14	-0.08	0.39	0.57
31	2.2	-2.84	49.87	0.47	50.17	-0.23	-0.08	0.35	0.49
32	1.53	-2.46	44.1	0.32	49.71	0.26	0.13	0.37	0.39
33	1.97	-0.67	53.14	0.46	52.2	-0.03	-0.01	0.34	0.49
34	2.66	0.74	64.06	0.7	42.05	-0.32	-0.14	0.33	0.55
35	2.5	-1.72	58.97	0.65	26.62	-0.23	-0.1	0.41	0.47

Table 6 (continued)

Sample no	SAR (meq/L)	RSC (meq/L)	PI (meq/L)	KR (meq/L)	MAR (meq/L)	Indices of base exchange	Gibbs ratio I (meq/L)	Gibbs ratio II (meq/L)	%Na (meq/L)
						CAI-I (meq/L)	CAI-II (meq/L)		
36	2.56	0.54	63.17	0.67	33.98	-0.25	-0.11	0.35	0.5
37	3.5	1.27	73.6	1.04	43.19	-0.46	-0.22	0.37	0.65
38	1.89	-1.53	52.06	0.45	62.23	-0.2	-0.06	0.32	0.54
39	1.82	-1.41	45.17	0.35	62.23	-0.09	-0.03	0.27	0.48
40	2.35	-2.69	51.35	0.51	40.93	0.07	0.05	0.43	0.46
41	2.27	-1.44	55.9	0.55	54.83	0.14	0.09	0.44	0.55
42	2.13	-1.53	53.92	0.51	28.84	0.17	0.11	0.44	0.42
43	1.14	-7.72	31.52	0.2	82.34	0.51	0.32	0.44	0.53
44	2.4	-0.7	59.29	0.61	42.82	0.09	0.06	0.44	0.52
45	2.65	-0.59	63.23	0.72	38.84	-0.05	-0.03	0.44	0.54
46	2.8	0.34	65.85	0.77	48.22	-0.28	-0.13	0.37	0.6
min	1.14	-8.66	30.39	0.2	8.38	-1.56	-0.31	0.25	0.25
max	4.171	1.38	73.6	1.04	82.34	0.51	0.32	0.5	0.65
Avg	2.23	-2.35	51.74	0.52	41	-0.06	0.01	0.38	0.45
Stdev	0.63	2.57	11.49	0.22	14.18	0.39	0.15	0.06	0.1

**Fig. 4** Wilcox diagram for water categorization using EC and % Na

$$KR = \frac{Na^+}{(Ca^{2+} + Mg^{2+})}$$

where the ion concentration is measured in mg/L. KR values in the research area ranged from 0.20 to 1.04 mg/L (Table 6). A KR of < 1 is ideal for farming, while a KR of > 1 is not. Only 98% of groundwater sources are appropriate for irrigation, with only 2% of samples not fit for agricultural production (Table 7).

Magnesium adsorption ratio (MAR)

One of the most vital qualitative parameters in establishing the excellence of irrigation water is the magnesium content. Most waters have a balance of calcium and magnesium. As soils become more saline, more magnesium in water will negatively impact crop production (Szabolcs 1964; Zhang et al. 2020). The magnesium adsorption ratio ranges from 8.38 to 82.34 meq/L, in the current investigation (Table 6). The remaining 22% of groundwater samples were deemed inappropriate, with 78% of samples falling below the permissible standard (Table 7). This is due to the fact that when the magnesium adsorption ratio exceeds 50%, it is damaging to the soil.

$$MAR = \frac{Mg^{2+}}{(Ca^{2+} + Mg^{2+})} \times 100$$

Table 7 Water compatibility for irrigation is used to classify the quality of water

Parameter	Water quality	Ranges	Sample numbers	% of groundwater samples
SAR (Richards 1954)	Excellent	0–10	1–46	100%
	Good	10–18		
	Doubtful	18–26		
	Unsuitable	> 26		
EC	Excellent	< 250	–	78%
	Good	250–750		
	Permissible	750–2000		
	Doubtful	2000–3000		
TDS (Freeze and Cherry 1979)	Unsafe	> 3000	–	2%
	Fresh	< 1000		
	Brackish	> 1000		
	Saline	> 10,000		
TH (Sawyer and McCarty 1967)	Braine	> 100,000	–	22%
	Safe	< 75		
	Moderately hard	75–150		
	Hard	150–300		
RSC (Eaton 1950)	Very hard	> 300	37	2%
	Good	< 1.25		
	Doubtful	1.25–2.50		
	Unsuitable	> 2.50		
%Na (Wilcox 1955)	Excellent	< 20	1–4, 6–36, 38–46	93%
	Good	20–40		
	Permissible	40–60		
	Doubtful	60–80		
PI (Doneen 1964)	Unsuitable	> 80	–	7%
	Unsuitable	< 25 Class-III		
	Good	25–75 Class-II		
	Excellent	> 75 Class-I		
KR (Kelley 1940)	Suitable	< 1	1–36, 38–46	98%
	Unsuitable	> 1		
MAR (Szabolcs 1964)	Suitable	< 50	1–12, 14, 18–23, 25–30, 32, 34–37, 40, 42, 44–46	78%
	Unsuitable	> 50		

Gibbs diagram

Gibbs (1970) provided a schematic for interpreting the primary ion chemistry mechanism in groundwater samples (Figs. 6 and 7). Precipitation dominance, evaporation dominance, and rock dominance are the three separate fields explained in Gibbs graphic. The following formula was used to determine Gibbs ratio, and all ions' concentrations were measured in meq/L. Between Gibbs ratio (cation or anion) and TDS, Gibbs diagram was plotted. Gibbs ratio 1 values vary between 0.01 to 0.66 in the current study region, with a mean of 0.33. Gibbs ratio 2 values ranged from 0.04 to 0.45 in the current study area, with a mean of 0.143 (Table 6).

$$\text{Gibbs ratio 1 (anion)} = \frac{(Cl^- + NO_3^-)}{(Cl^- + NO_3^- + HCO_3^-)}$$

$$\text{Gibbs ratio 2 (cation)} = \frac{(Na^+ + K^+)}{(Na^+ + K^+ + Ca^{2+})}$$

Conclusion

The current study examines groundwater quality in the Pandameru River basin, a region predominantly characterized by hard rocks, which is essential for drinking and irrigation purposes. Parameters were compared to WHO and BIS criteria for drinking water acceptability, and graphical approaches were utilized to evaluate groundwater quality

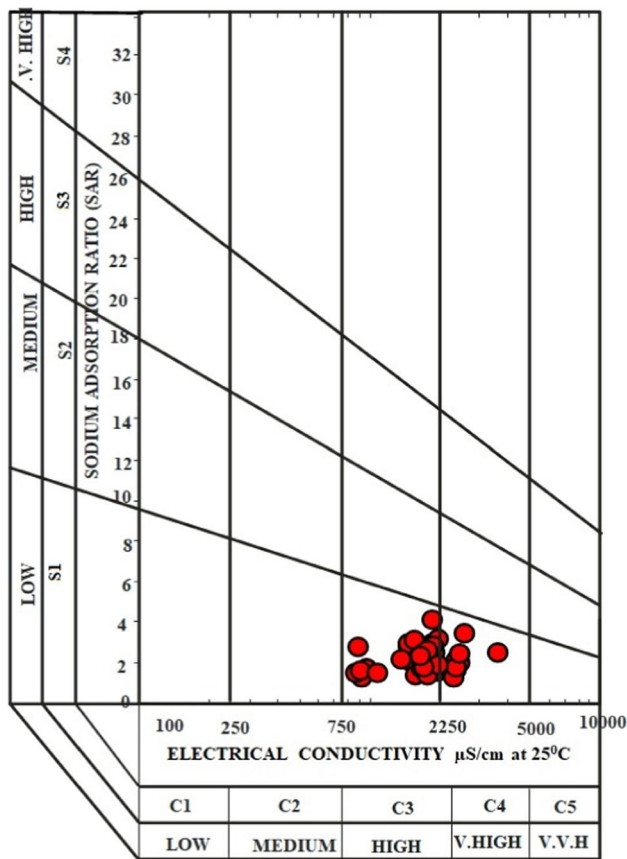


Fig. 5 Irrigation water quality classification based on salinity and sodium hazard (following the US Salinity Laboratory, 1954)

for irrigation. The following conclusions were drawn from the analysis:

The groundwater in the study area is of slightly alkaline quality. The distribution of major ions in the groundwater is found in the following order: $\text{Ca}^{2+} > \text{Na}^+ > \text{Mg}^{2+} > \text{K}^+$ and

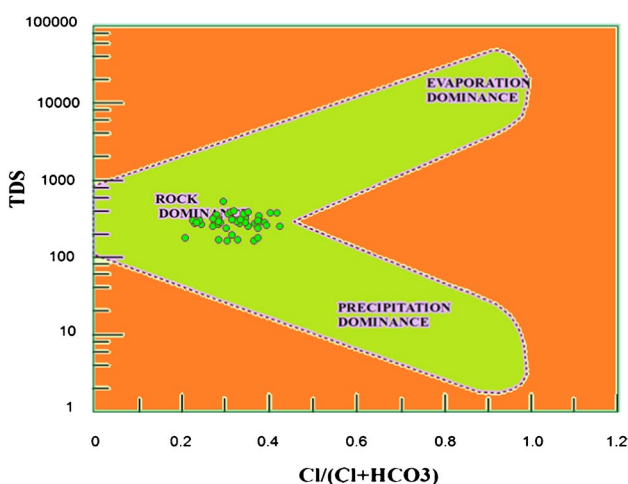


Fig. 6 Gibbs plot of the ratio of major anions vs. TDS

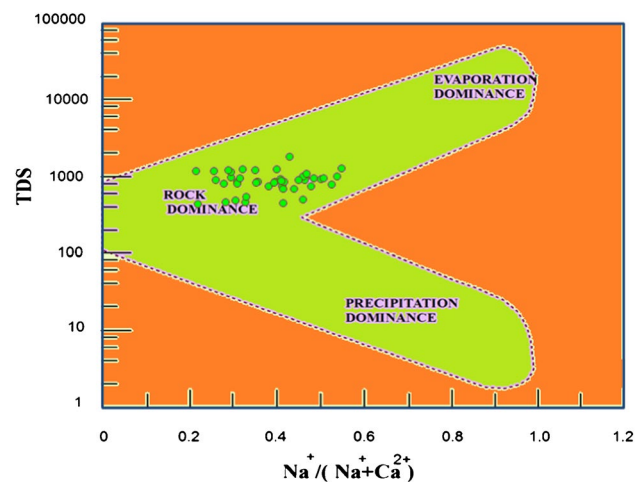


Fig. 7 Gibbs plot of the ratio of major cations vs. TDS

$\text{HCO}_3^- > \text{Cl}^- > \text{SO}_4^{2-} > \text{NO}_3^-$. The results demonstrate that Ca^{2+} , Mg^{2+} , HCO_3^- , and Cl^- concentrations in the study region exceed the (World Health Organization 2012) maximum permissible limit. According to the WQI classification, only 77% of groundwater samples are absolutely fit for drinking, while the remaining 23% of groundwater samples are unfit for drinking. According to the USSL diagram, 83% of groundwater samples fall into the C3S1 category, which denotes water with a low salt concentration and low sodium content, making it ideal for irrigation, while the remaining 17% of groundwater samples fall into the C4S1 category, which denotes water with very saline conditions and low sodium content, making it unsuitable for irrigation. According to Wilcox classification, 78% of groundwater samples are good for cultivation, whereas 20% and 2% are doubtful, and unsuitable, respectively. Furthermore, based on %Na, 100% of subsurface water samples are within permitted limits for agricultural usage. According to the permeability index, the water in the research area is suitable for agriculture.

The study identified high electrical conductivity and elevated concentrations of Ca^{2+} , Na^+ , HCO_3^- , Cl^- , and SO_4^{2-} in groundwater, potentially attributed to mineral disintegration and human activities such as wastewater interaction, intensive fertilizer use, and agricultural practices. The research estimated the potable water quality index (WQI) and mapped the spatial distribution of chemical constituents in the Pandameru River basin using advanced tools like GIS and remote sensing. The maps offer crucial insights for the public to comprehend groundwater contamination. Limited use of groundwater for irrigation was identified in specific areas, highlighting the need for the introduction of salt-tolerant crops and enhanced drainage systems. In order to improve water quality, it is advised that the government collaborates with NGOs to implement

additional rainwater harvesting structures, including infiltration canals, control barriers, replenishment trenches, farming ponds, and artificial recharge structures.

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Author contribution P. Ravi Kumar: manuscript preparation, sample collection, and sample analysis.

S. Srinivasa Gowd: corresponding author, manuscript corrections.

C. Krupavathi: sample collection and analysis.

B. N. Anusha: remote sensing and GIS mapping.

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